

Acquired Immunodeficiency Syndrome (AIDS)

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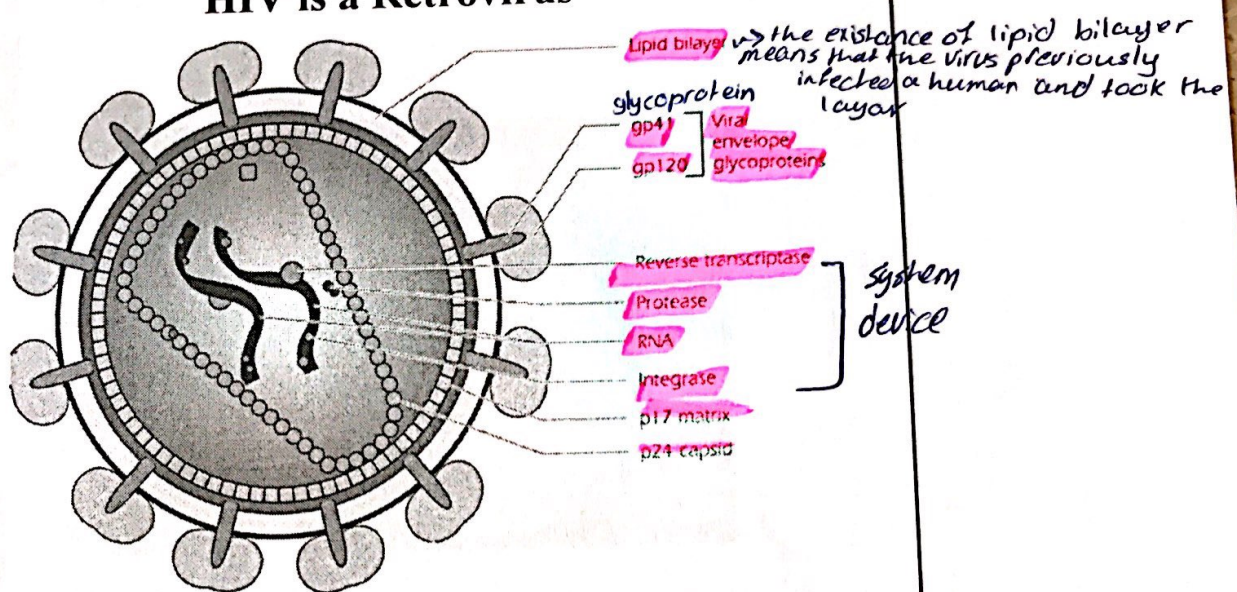
* the HIV virus was discovered in 1981, but did it exist before that? No, from the moment it was discovered it has been spreading ~~wildly~~ wildly

Acquired Immunodeficiency Syndrome (AIDS)

History

- ◆ 1981: First reports of "Acquired Immuno-deficiency Syndrome" in Los Angeles.
- ◆ 1983: Virus first isolated in France (LAV).
- ◆ 1984: Virus isolated in the U.S. (called HTLV-III and AIDS-Related Virus)
- ◆ 1986: Consensus name Human Immunodeficiency Virus (HIV-1).
- ◆ 2009: AIDS patients 35 millions
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- ◆ 2010: AIDS patients 36 millions
- ◆ 2011: AIDS patients 35.7 millions
- ◆ 2012: AIDS patients 35.3 millions
- ◆ 2015: AIDS patients 38.2 millions

Structure of the Human Immunodeficiency Virus HIV is a Retrovirus



The viral structure

• HIV is spherical in shape, 1/10,000 mm diameter.

• The viral envelope is composed of:

- a. two layers of fatty molecules, taken from the membrane of a human cell.
- b. Embedded Env proteins from the host cell (gp120 cap and gp41)

lipid bilayer

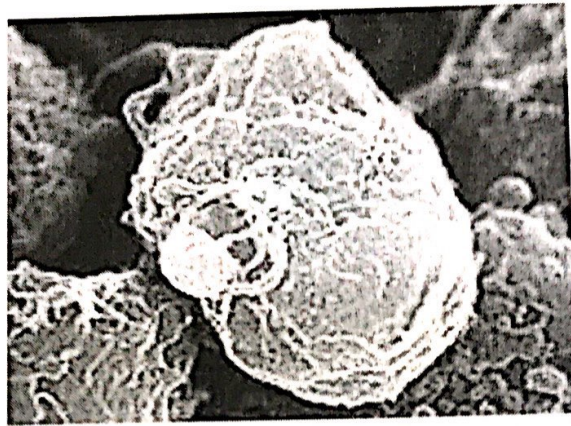
The core

The capsid: 2,000 copies of the viral protein, p24. It surrounds two single strands of HIV RNA.

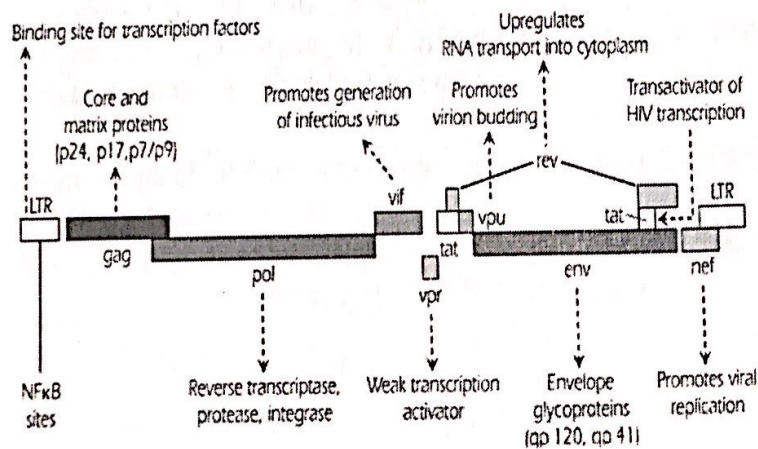
Three enzymes: reverse transcriptase, integrase, and protease.

p17 matrix protein, lies between the viral core and the viral envelope.

HIV virus in electron microscopy



The genes and proteins of HIV



■ The genes of HIV

- 3 structural genes (gag, pol, and env).
env gene codes for a protein called gp160 that is broken down by a viral enzyme to form gp120 and gp41
gag codes for core and matrix proteins P24 and p17.
pol codes for reverse transcriptase, protease and integrase.
- HIV has six regulatory genes (tat, rev, nef, vif, vpr, and vpu) that contain information needed to produce proteins that control the ability of HIV to infect a cell, produce new copies of virus, or cause disease.
- **vif protein:**
 interacts with an antiviral defense protein (the cyt deaminase) in host cells, promote its degradation, enhancing HIV replication.
- The ends of each strand of HIV RNA contain an RNA sequence called the long terminal repeat (LTR). control production of new viruses

Life Cycle of HIV

- ① 1. Attachment: Macrophage tropic HIV: In MALT, Virus gp120 binds to surface molecule CD4 of macrophages and dendritic cells and CCR5 Coreceptor. CCR5 is the major coreceptor for the establishment of primary infection.
2. Lymphotropic HIV: Virus gp120 binds to surface molecule CD4 of Lymphocytes and CCR4 Coreceptor.
- ② 3. Fusion: Viral envelope fuses with cell membrane, releasing RNA contents into the cell.
- ③ 4. Reverse Transcription: Reverse transcriptase makes DNA
 RNA -----> DNA

or lymphocytes
 (T_H cells)
 bind with CCR4

Life Cycle of HIV

④
5. **Integration:** Viral DNA is inserted into host cell chromosome by unique enzyme *integrase*. Integrated viral DNA may remain latent for years and is called a provirus.

6. Replication:

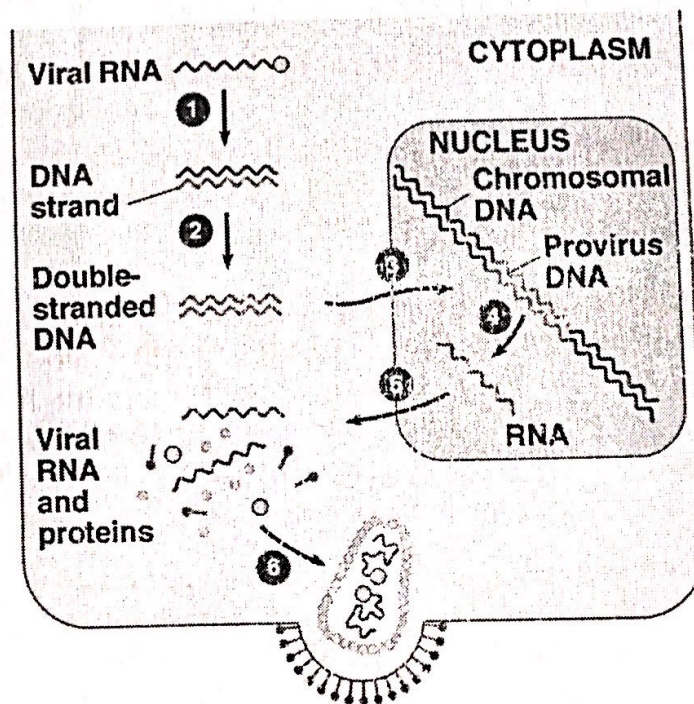
⑤ Viral genome is replicated.

⑥

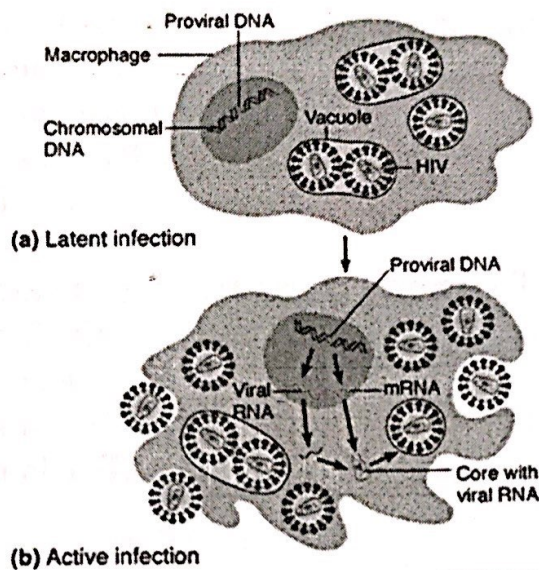
7. Assembly: New viruses are made and released.

⑦ Lyses of CD4⁺ T_H cells

HIV Life Cycle: Reverse Transcriptase Converts RNA into DNA



HIV Life Cycle: Latent versus Active Infection in Macrophages



Acute Infection

- Asymptomatic, Flu-like, starts 2-4 weeks and lasts 1-2 weeks

- Infection and death of many CD4⁺ including memory cells in mucosal lymphoid tissue.

Predominate location: lymph nodes (lymphadenopathy)

Graded decline in CD4⁺, Reverse ratio CD4/CD8=1/2

HIV cytopathic effect, Cell lysis, CTL attack, Apoptosis

- Uninfected cells are also lysed: gp120 protein will be released and bound to T cells to become foreign

* uninfected cells are more than infected cells but both are lysed

- Tat protein: interact with regulatory proteins and interfere with Cytokine synthesis.

Tat enters the nucleus of infected cells and escape from plasma mem to enter neighboring uninfected cells

* immune system do make antibodies against HIV's cell surface proteins but they don't work

- Infected Mac and Dendritic cells disseminate the virus to lymphoid tissue -----Viremia

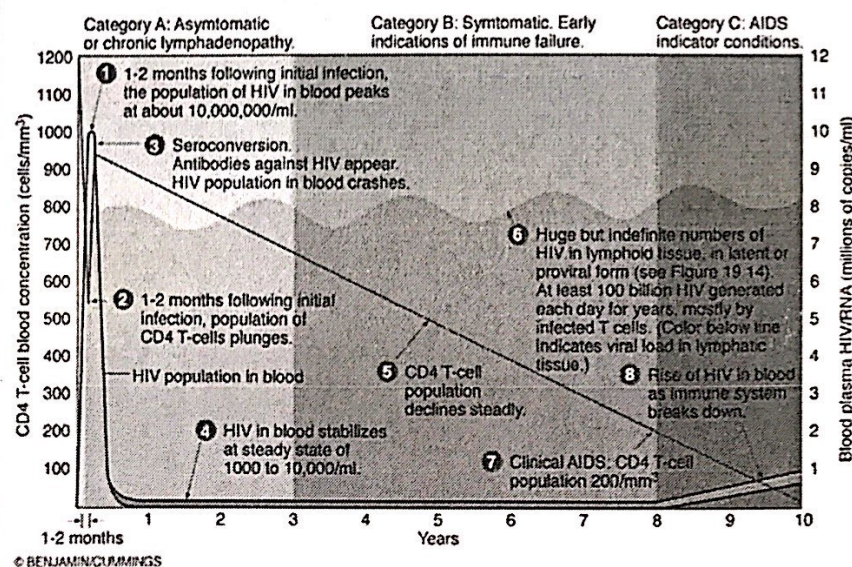
Mac express lower levels of CD4 than Th but express CCR5, resistant to cytopathic effect---act as reservoir

- Immune response:

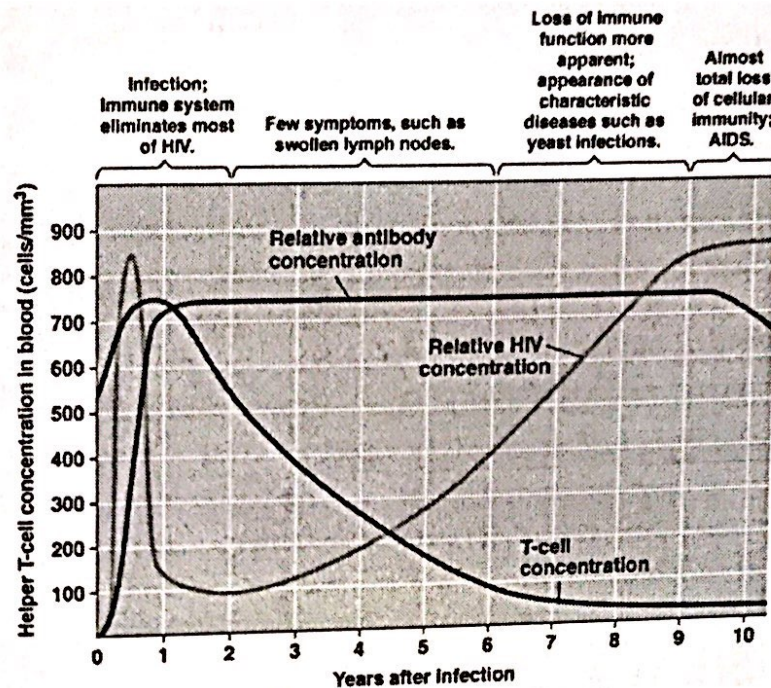
Anti-HIV Abs (Seroconversion) and HIV-specific CTL

- Expression of the activation markers CD38 and HLA-DR on T cells: the best correlates of disease progression.

- Elevation of the pro-inflammatory cytokines including TNF-alpha, IL-6 and IL-1beta.



Antibody Levels, T Cell Counts, and HIV Concentration After Infection



* provirus $\rightarrow$ activated $\rightarrow$ AIDS

Chronic latent infection

* Asymptomatic, may last 15 years

* infectious

* low level of viral replication

HIV never truly latent.

* The number of infected cells in blood is very low
Less than 1% of CD4+ cells are infected.

AIDS CRISES

Blood CD4+ cells: $<200\text{cell/mm}^3$

1. **Unusual malignancies:**

**Sarcomas (Kaposi), Carcinomas (cervical),
Lymphomas (Burkitt's and CNS).**

2. **Opportunistic Infections:**

**Bac (Mycobacteria), viral (CMV), Parasites
(Toxoplasma) and fungal (Candida)**

3. **CNS abnormalities:**

Memory loss, motor problems

** even when HIV is still a provirus it's infectious*

Transmission of AIDS (Worldwide)

1. **Sexual contact with infected individual**

2. **Sharing of unsterilized needles by
intravenous drug users and unsafe
medical practices**

3. **Transfusions and Blood Products**

4. **Mother to Infant (Perinatal)**

AIDS Associated Disease Categories

2. Respiratory: 70% of AIDS patients develop serious respiratory problems.

Partial list of respiratory problems associated with AIDS:

- Bronchitis
- Pneumonia
- Tuberculosis
- Lung cancer
- Sinusitis
- Pneumonitis

* how do the virus escape the immune system?

↳ hyper mutations
The genes of the virus are susceptible to mutations ex. HIV in Jordan is different than HIV in the US
specially in gp120
keep changing so the immune system won't recognize it
(envelope proteins are very susceptible to mutations)

Escape mechanisms how the virus escape i.s

- Lack of fidelity of RT and RNA polymerase: Mutation in gp120 one of the most variable component varies from the same individual (everyday). Mutations are frequent due to HIV's error prone replication and the structural flexibility of the viral envelope (Env).
A region in gp120 called V3 loop is the most antigenically variable: different in the same individual
- 2- ■ Covers itself with CD59 and DAF to protect from complement
- 3- ■ DNA in provirus is nonantigenic so it hide itself
- 4- ■ In contrast to activated T cells, Naïve T cells resist HIV infection: Naïve cells have cyt deaminase that mutate HIV and inhibit it. * cyt → cytidine (enzyme)
Activated T cells have inactive E, Vif protein bind the E and promote its degradation
- 5- ■ HIV also evades antibody by coating itself with sugars. (polysaccharide) which leads to glycosylation
 - a. By adding asparagine-linked glycosylations during processing of envelope proteins (gp160),
 - b. the virus is able to produce a 'glycan shield', thus preventing antibody binding.
 - c. point mutations, insertions, and deletions can alter the positioning of the sugars, resulting in an ever-changing disguise